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ROLL NO:-49

EXPERIMENTS :-1,2,3,4,12

Insertion Sort:-

Code:-

```
#include <stdio.h>
void insertionSort(int arr[], int n) {
 int i, key, j;
 for (i = 1; i < n; i++) {
 key = arr[i];
 j = i - 1;
 while (j >= 0 && arr[j] > key) {
 arr[j + 1] = arr[j];
 j = j - 1;
 }
 arr[j + 1] = key;
 }
}
int main() {
 int n, i;
 printf("Enter number of elements in the array: ");
 scanf("%d", &n);
 int arr[n];
```

```
 printf("Enter %d integers: ", n);
 for (i = 0; i < n; i++)
 scanf("%d", &arr[i]);
 insertionSort(arr, n);
 printf("Sorted array: ");
 for (i = 0; i < n; i++)
 printf("%d ", arr[i]);
 printf("\n");
 return 0;
}
```

OUTPUT:-

```
Output
/tmp/03ySHJSD1e.o
Enter number of elements in the array: 5
Enter 5 integers: 2 13 5 18 14
Sorted array: 2 5 13 14 18
```

Selection Sort:

Code:

```
#include <stdio.h>
void swap(int *xp, int *yp)
{
        int temp = *xp;
        *xp = *yp;
```

```
	*yp = temp;
}
void selectionSort(int arr[], int n)
{
	int i, j, min_idx;
	for (i = 0; i < n-1; i++)
	{
		min_idx = i;
		for (j = i+1; j < n; j++)
		if (arr[j] < arr[min_idx])
			min_idx = j;
		if(min_idx != i)
			swap(&arr[min_idx], &arr[i]);
	}
}
void printArray(int arr[], int size)
{
	int i;
	for (i=0; i < size; i++)
		printf("%d ", arr[i]);
	printf("\n");
}
int main()
{
	int arr[] = {64, 25, 12, 22, 11};
```

```
        int n = sizeof(arr)/sizeof(arr[0]);
        selectionSort(arr, n);
        printf("Sorted array: \n");
        printArray(arr, n);
        return 0;
}
```

Output:

Sorted array:

11 12 22 25 64

BINARY SEARCH :-

CODE:-

```
#include<stdio.h>
int main()
{
 int i,low,high,mid,n,key;
 int a[100];
 printf("Enter the size of array:\n");
 scanf("%d",&n);
 printf("Enter sorted elements:\n");
 for(i=1;i<=n;i++){
  scanf("%d",&a[i]);
```

```
 }
 printf("Enter the key:\n");
 scanf("%d",&key);
 low=1;
 high=n;
 while(low<=high){
  mid=(low+high)/2;
  if(a[mid]==key){
   printf("element %d found at position %d",key,mid);
   break;
  }
  else if(a[mid]<key){
   low=mid-1;
  }
  else{
   high=mid+1;
  }
 }
 return 0;
}
```

OUTPUT:-

```
/tmp/t6Uld6x4e6.o
Enter the size of array:
4
Enter sorted elements:
12
14
10
5
Enter the key:
14
element 14 found at position 2
```

Naive string Algorithm

code:-

```
#include <stdio.h>
#include <string.h>
int main (){

  char txt[100], pat[100];

  int M, N;

  printf("Enter the text: ");

  scanf("%s", txt);

  printf("Enter the pattern: ");
```

```
  scanf("%s", pat);

  M = strlen(pat);
  N = strlen(txt);

  for (int i = 0; i <= N - M; i++){
    int j;

    for (j = 0; j < M; j++)

      if (txt[i + j] != pat[j])
        break;

    if (j == M)
      printf ("Pattern matches at index %d \n", i);
  }

  return 0;
}
```

OUTPUT:-

```
/tmp/Hu2Pccs1bT.o
Enter the text: wearestudentofvcet
Enter the pattern: vcet
Pattern matches at index 14
```

QUICK SORT :

CODE:-

```
#include<stdio.h>
void quicksort(int a[50],int first,int last){
  int i, j, pivot, temp;
  if(first<last){
    pivot=first;
    i=first;
    j=last;
    while(i<j){
      while(a[i]<=a[pivot]&&i<last)
        i++;
      while(a[j]>a[pivot])
        j--;
      if(i<j){
        temp=a[i];
        a[i]=a[j];
        a[j]=temp;
      }
    }
    temp=a[pivot];
    a[pivot]=a[j];
    a[j]=temp;
```

```
    quicksort(a,first,j-1);
    quicksort(a,j+1,last);
  }
}

int main(){
  int i, n, a[50];
  printf("Enter no of elements: ");
  scanf("%d",&n);
  printf("Enter %d elements: ", n);
  for(i=0;i<n;i++)
    scanf("%d",&a[i]);
  quicksort(a,0,n-1);
  printf("sorted array: ");
  for(i=0;i<n;i++)
    printf(" %d",a[i]);
  return 0;
}
```

Output:

```
Enter no of elements: 4
Enter 4 elements: 44
11
88
33
sorted array:  11 33 44 88
```